

JPHG01 Temperature and thermometer — Quiz 2

S3 Physics · 3 multiple-choice items

Q1 · JPHG01.2 · Applied

A thermometer makes use of the resistance R of a platinum wire wrapped around a ceramic core to measure temperature T . The thermometer is calibrated for the two fixed points. R is assumed to change linearly with T and the solid line in the figure is used to give the reading of the thermometer. The dotted line shows the actual variation of R with T . If the thermometer gives a reading of $120\text{ }^{\circ}\text{C}$ in a measurement, which of the following statements is correct?

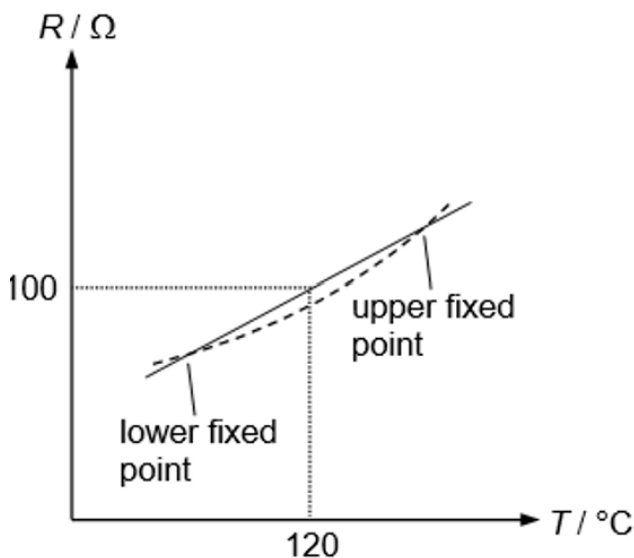


Fig · R–T graph of the platinum resistance thermometer

- A. The actual temperature is lower than $120\text{ }^{\circ}\text{C}$.
- B. The actual temperature is higher than $120\text{ }^{\circ}\text{C}$.
- C. The resistance of the platinum wire is smaller than $100\text{ }\Omega$.
- D. The resistance of the platinum wire is larger than $100\text{ }\Omega$.

Q2 · JPHG01.3 · Foundation

A mercury-in-glass thermometer will response to temperature change more quickly if

- A. the inner diameter of the tube is reduced.
- B. the thickness of the wall is reduced.
- C. the size of the bulb is increased.
- D. the length of the tube is increased.

Q3 · JPHG01.3 · Foundation

Which of the following methods can improve the sensitivity of a liquid-in-glass thermometer?

- (1) Reduce the diameter of the glass tube.
- (2) Increase the length of the glass tube.
- (3) Use a liquid which expands less for the same temperature increase.

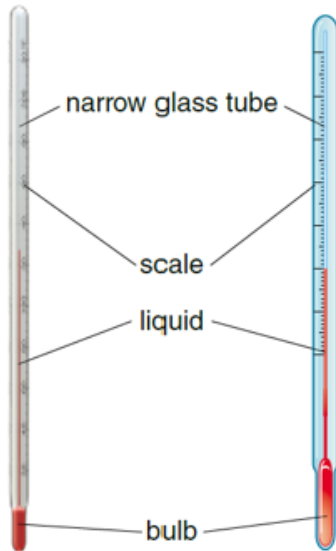


Fig · Structure of a liquid-in-glass thermometer

- A.** (1) only
- B.** (2) only
- C.** (1) and (3) only
- D.** (2) and (3) only

JPHG01 Temperature and thermometer — Quiz 2 · Answer key

Teacher page · 3 items

Q1 · JPHG01.2

Answer: B. The actual temperature is higher than 120 °C.

Guide: The reading 120 °C comes from the solid (linear) scale, so the measured resistance is 100 Ω . On the actual (dotted) curve, $R = 100 \Omega$ occurs at a temperature higher than 120 °C.

Q2 · JPHG01.3

Answer: B. the thickness of the wall is reduced.

Guide: A thinner wall lets heat reach the mercury faster, so the thermometer responds more quickly. A narrower tube or larger bulb changes sensitivity, not response time.

Q3 · JPHG01.3

Answer: A. (1) only

Guide: A narrower tube gives a larger column rise for the same volume expansion. A longer tube mainly affects range. A liquid that expands less makes the column rise less, so sensitivity falls.